

A Synchronized-Lapse Cosmology: Real Thermal History inside a Temporary Metric Lapse

Formalization of a reconciled Day-4 creation model

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Status: model definition. No empirical confirmation is claimed.

Abstract

A cosmological model is stated in which the Earth-frame creation days of Genesis 1 are terrestrial proper-time intervals, while the extraterrestrial region executes the standard hot thermal history during the Day-4 interval under a temporary metric lapse. The lapse multiplies proper time. It is not a post-hoc rescaling of photon frequency. Redshift, supernova time dilation, intermediate absorption, lensing delays, light-element yields, and the acoustic scale of the microwave background are inherited from a single scale factor in the cosmic thread. Design is located in the initial data and in the existence and duration of the lapse. At synchronization the lapse flattens, the inbound light cone is already populated, and subsequent evolution proceeds under ordinary law. The synchronization condition is not a block universe: it does not pre-write outcomes after the barrier. Failure conditions are stated. The construction is silent on terrestrial radiometric and stratigraphic chronology.

1. Problem

Genesis 1 places the terrestrial frame first and the luminaries on the fourth day, “to give light upon the earth.” A recent Earth-frame chronology then has to account for an inbound light cone that already contains sources whose emission, in standard cosmology, lies millions to billions of years down the past null geodesic. The difficulty is not a single travel time. Spectroscopic redshift, the stretch of Type Ia light curves, the Lyman-alpha forest, gravitational-lens delays, primordial abundances, and the angular scale of cosmic-microwave-background anisotropies are correlated. They are one expansion history sampled at different events. A translation layer that rescales frequencies after emission must reproduce that joint structure or become a second, hand-built copy of the history it was meant to avoid.

This paper formalizes a reconciliation that does not replace the thermal history. The history runs. It runs in cosmic proper time, inside an interval whose terrestrial proper duration is one day, because a metric lapse suppresses the Earth clock only during that interval. The construction is a definition of the model, not a claim that the lapse has been derived from a solved initial-value problem.

2. Relation to prior proposals

The model sits in a known family and differs from each member in a stated way.

Gravitational time-dilation and white-hole constructions (Humphreys; Hartnett) already use unequal clocks so that cosmic process can be long in one frame and short in another. The present model keeps that idea and requires that the fast region contain the standard thermal history rather than a truncated or projected substitute. It also requires the lapse to be negligible after synchronization, so that present-day clock comparisons and microwave isotropy are not asked to absorb a residual anomaly.

Anisotropic synchrony (Lisle) keeps one-way inbound light instantaneous by a convention on clock synchronization, with the round-trip speed held at c . The present model does not adopt that convention. Inbound photons travel on ordinary null geodesics of the lapse-modified metric. Their emission events lie in the cosmic past of the fast thread.

Mature-creation and functional-maturity accounts instantiate a working cosmos without the processes that would have produced it. The present model moves maturity from the sky to the initial data. Stars, remnants, abundances, and the microwave background are products of the run. They are designed only in the sense that the initial data and the lapse were chosen so the run would complete inside one terrestrial day.

A frequency-translation layer between a hyper-accelerated compile step and local optics is rejected. No such layer is an independent degree of freedom in the model.

3. Postulates

The model is fixed by five postulates. Anything not entailed by them is outside the model.

P1. Reference clock. The creation days are intervals of proper time on the terrestrial contour. No other clock defines the day.

P2. Initial data. At the start of the Day-4 interval the cosmic region is given constants, a baryon-to-photon ratio, a fluctuation amplitude, a spatial curvature, and a stress-energy inventory, including whatever component will cluster and whatever component will dominate late. These are boundary conditions, not outputs.

P3. Temporary lapse. During the Day-4 interval a metric lapse suppresses terrestrial proper time relative to cosmic proper time. Cosmic proper time advances through recombination, reionization, and the stellar history while terrestrial proper time advances one day. The lapse multiplies proper time. It does not multiply photon energy by a separate function of distance.

P4. Ordinary process in the fast region. Within the cosmic region the field equations and the particle processes are the ordinary ones. Light-element yields, recombination, structure growth, and stellar evolution are computed as in the standard thermal history, on cosmic proper time.

P5. Synchronization, not a block. At the end of the Day-4 interval the lapse flattens to the ordinary cosmological value, up to local potentials. The inbound light cone of the terrestrial frame is already populated by emissions from the fast thread. Subsequent evolution uses one shared rate. The barrier does not assert that later events are already actual.

4. Metric and lapse

Write the spacetime in 3+1 form. With lapse α , shift β^i , and spatial metric γ_{ij} ,

$$ds^2 = -\alpha^2 c^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt). \quad (1)$$

Proper time along a worldline at rest in these coordinates satisfies $d\tau = \alpha dt$. Let τ_E be proper time on the terrestrial contour and τ_C proper time on a comoving worldline in the cosmic region. Over the Day-4 coordinate interval I_4 ,

$$\int_{I_4} \alpha_E dt = 1 \text{ day}, \quad \int_{I_4} \alpha_C dt = \tau_{\text{thermal}},$$

(2)

where τ_{thermal} is the cosmic proper time from the hot early state to synchronization, of order the conventional 13.8 Gyr if the fast thread is the standard history. Equation (2) is the content of P3. It does not by itself exhibit a stable solution of the Einstein equation with this lapse profile. Constructing $\alpha(x, t)$ such that (2) holds, the dominant energy condition is respected outside the stipulated initial data, and the terrestrial region remains intact, is an obligation of the model, not a theorem of this paper.

After the synchronization time t_s ,

$$\alpha_E / \alpha_C \rightarrow 1, \quad (3)$$

up to the ordinary solar-system potential and the ordinary cosmological lapse of the post-barrier expansion. A lapse ratio that remained of order $\tau_{\text{thermal}} / 1$ day after t_s is forbidden by P5. Such a residual would appear in present-day clock comparisons and, unless finely screened, in the isotropy of the microwave background.

Photon energy is not assigned an extra map. Along a null geodesic, frequency shift relative to emitters and observers at rest in the slicing comes from the expansion of γ_{ij} and from the integrated lapse and shift. In the cosmic region after the early transient, $\gamma_{ij} = a(\tau)^2 \delta_{ij}$ in the usual way, and the redshift between emission at τ_e and observation after synchronization is

$$1 + z = a(\tau_o) / a(\tau_e), \quad (4)$$

provided the post-barrier potential difference does not add a comparable shift. Equation (4) is the standard relation. The model earns it by refusing a separate frequency layer, not by altering it.

5. Thermal history in the fast thread

On cosmic proper time the thread executes the following sequence. The items are outputs of P2 and P4, not independent stipulations.

Nucleosynthesis. The baryon-to-photon ratio in the initial data fixes the helium and deuterium yields. Agreement with observed abundances is then a constraint on that initial datum, as in the standard account, not a separate miracle at synchronization.

Recombination. The baryon-photon fluid releases a blackbody at the usual epoch. The sound horizon r_s at last scattering is a length computed in the fast thread. After synchronization it subtends the observed acoustic angle only if the angular-diameter distance is the distance belonging to the same $a(\tau)$. The monopole temperature scales as $1/a$ and is the observed 2.725 K at the barrier if $a(\tau_s)$ is the observed scale factor. Polarization and the damping tail are properties of that fluid. They are not a deployment footprint.

Structure. Growth in the fast thread requires a clustering component present while it grows. Rotation curves, cluster dynamics, lensing, and the acoustic-peak ratios are outputs of that component. The model does not rename the component; it requires that whatever plays the role act during the run. A smooth late component in the committed stress-energy produces the magnitude-redshift relation of supernovae that explode at the corresponding a . Acceleration is late-time behavior already on the cone, not a pressure term painted at t_s .

Stars. Ignition, main-sequence lifetimes, and remnants occur in cosmic proper time. A Type Ia light curve stretches by $1 + z$ because the explosion clock and the photon stretch share $a(\tau)$. Stellar isochrones and white-dwarf sequences in external galaxies are the same fact read in remnants. The inbound cone is full at t_s because these events have already happened in τ_C , not because an image of them was compiled.

6. Synchronization

Synchronization is the surface $\tau_E = \text{end of Day 4}$, equivalently $t = t_s$, on which three conditions hold together.

S1. Equation (3) is in force.

S2. Every past-directed null geodesic from the terrestrial contour back to the emission epoch of an observed source terminates on an event inside the fast thread.

S3. The macroscopic boundary data on the surface are fixed. The future domain of dependence is not declared already actual.

S3 is the exclusion of a block universe. A block reading, on which past, present, and future are equally real, may be laid over S1–S2. It is not entailed. A growing-block reading keeps the completed cone fixed and leaves the future open. A presentist reading treats fast-thread events as having happened, in τ_C , and as now past. The lapse explains the fit inside one terrestrial day. Neither indeterminacy after the barrier nor ordinary causal openness is in tension with S1–S3. Both are in tension only with the optional block gloss.

Horizon and flatness are not solved by synchronization. They are either set in the initial data of P2 or produced by a brief early acceleration inside the fast thread. Both are available. The model does not choose between them here.

7. Inherited observables

Table 1 lists the observables the model inherits from the fast thread, and the condition under which the inheritance holds. Where the condition fails, the model fails with it. The table is not a fit to data. It is a map from postulate to observable.

Observable	Source in the model	Inheritance condition
Redshift–distance	$a(\tau)$ along the geodesic	No extra frequency map; eq. (4)
SN Ia stretch	Emitter proper time $\times a(\tau)$	Stretch equals $1 + z$
Lyman-alpha forest	Intervening gas at intermediate a	Geodesics cross the fast thread
Lens time delays	Path difference in the metric	No shear that detunes delays from z
CMB monopole	Recombination radiation, cooled as $1/a$	Blackbody at 2.725 K at t_s
CMB acoustic scale	Sound horizon / angular-diameter distance	Same $a(\tau)$ fixes both
Light elements	Baryon loading in P2	Standard yields at that loading
Clustering probes	Component present during growth	Acts in the run, not after it
Late acceleration	Smooth stress-energy in P2	Supernovae at the corresponding a

Table 1. Observables inherited from the fast thread. Inheritance fails if the lapse shears the single-factor link between redshift and clock stretch, or if the thread is not allowed to contain the process named in the second column.

8. What is designed

Design in this model is exhausted by two acts. The first is the choice of initial data in P2: constants, baryon loading, fluctuation amplitude, curvature, and the stress-energy inventory. The second is the lapse of P3: that a suppression of the terrestrial clock exists, that it lasts one terrestrial day, and that it ends. Products of the run — ash, remnants, the blackbody, the web — are not separately designed. Calling them initialization artifacts is accurate only at one remove.

Fine-tuning is relocated, not removed. The fast thread must be aimed at the observed baryon density, fluctuation amplitude, and spatial flatness. The model does not explain those values. It declines to explain them twice, once in the initial data and again in a frequency layer.

9. Failure conditions

The model is falsified, or left without a realization, in any of the following cases.

F1. Lapse cannot start and stop. No solution of the field equations yields equation (2) during I_4 and equation (3) after t_s without a residual that violates microwave isotropy or present-day clock comparisons.

F2. Geodesic shear. The lapse profile shifts photon energy, or desynchronizes path delays, by a factor that is not a function of a alone, so that supernova stretch is not $1 + z$, or lens delays detach from the spectroscopic redshift.

F3. Truncated thread. The fast region is not permitted to contain nucleosynthesis, recombination, and stellar evolution. No post-hoc map restores the joint correlations of Table 1.

F4. Present-day anomaly. A discriminator that should be absent after synchronization is observed: a large local lapse, a directional microwave residual tied to the creation contour, or a cutoff in the stretch–redshift relation at a distance corresponding to t_s .

The model is not falsified by the sky looking old. On P3 and P4 the sky is old in τ_C . The claim is only that the Earth clock did not accumulate those years while the cone was being filled.

10. Scope

The model is cosmological. It does not speak to terrestrial parent–daughter ratios, ice, sediment, or magnetic stratigraphy. Those quantities live on the Earth clock after, and possibly before, the Day-4 interval. A lapse that suppresses only the terrestrial clock during Day 4 does not write them. Parent abundances required by the present radiogenic heat budget can sit in the Day-1 boundary data; daughter concordance cannot be derived from this paper’s postulates. Catastrophic rearrangement of the crust, if adopted, is a separate module and is not used here.

No numerical integration is offered. Order-of-magnitude cosmic time in equation (2) is the conventional thermal age, adopted so that the inherited observables match the standard history rather than a different one. A realization would have to produce $\alpha(x, t)$, check the isotropy integral along inbound geodesics, and show that the terrestrial contour survives the interval. Until that exists, the model is a conditional reconstruction: if such a lapse exists and ends, the optical and microwave evidence is the ordinary evidence, read on a clock that was slow while the cone filled.

11. Statement of the model

Earth is the reference clock. Day 1 sets the local contour and the cosmic initial data. Day 4 runs the standard thermal history in a cosmic region whose proper time is suppressed relative to Earth by a metric

lapse, for one terrestrial day. Photon frequency, light-curve stretch, and the acoustic scale come from that history's scale factor, not from a compile-time rescale. At the end of Day 4 the lapse flattens, the inbound cone is full, and both frames share one rate. The barrier synchronizes clocks. It does not freeze the future. Design sits in the initial data and in the lapse. The model fails if the lapse cannot be large only then, or if the thread is not the real history.

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